

MODULE 1 – INTRODUCTION

1. INTRODUCTION TO CYBER SPACE

Cyber space is the **virtual digital environment formed by interconnected computers**, servers, communication systems, networks, cloud platforms, and internet-based services.

It **allows users to access and exchange information globally in real time**. Example :- Online banking, e-commerce, social media, digital education, cloud storage, **and smart devices all operate within cyber space**.

Cyber space is not limited to the internet alone. It also includes:

- Private networks
- Cloud infrastructures
- Digital databases
- Mobile applications

Features of Cyber Space

- Global connectivity
- Fast communication
- Digital data sharing
- Remote access capability
- Real-time information exchange

Examples

- Online banking systems
- Social media platforms
- Cloud storage services
- E-commerce websites
- Smart home devices

2. INFORMATION SYSTEMS

An Information System (IS) is an **organized combination** of hardware, software, data, procedures, networks, and people that work together to collect, process, store, and distribute information.

Combination of hardware, software, data, and people used to process and manage information

Information systems help organizations:

- Manage operations
- Support decision-making
- Improve communication
- Store and analyze data

Without information systems, modern organizations cannot efficiently handle business activities or digital communication.

COMPONENTS OF INFORMATION SYSTEM

1. HARDWARE :- Physical devices used in computing.

Examples : Computers , Servers, Routers , Storage devices

2. SOFTWARE :- Programs and operating systems that perform tasks.
Examples: Windows , Linux , Database software

3. DATA :- Raw facts processed into useful information.

Examples: , Customer records , Transaction details

4. PROCEDURES :- Rules and instructions for operating systems securely.

5. PEOPLE :- Users, administrators, developers, and operators.

6. NETWORKS :- Communication channels connecting devices.
Examples: LAN , WAN ,Internet

TYPES OF INFORMATION SYSTEMS

- Transaction Processing System (TPS)
- Management Information System (MIS)
- Decision Support System (DSS)
- Executive Information System (EIS)

3. NEED FOR CYBER SECURITY

Cyber security refers to the **protection of computer systems**, networks, software, and data from cyber attacks, unauthorized access, theft, and damage.

Confidentiality **I**ntegrity **A**vailability

Cyber security mainly protects:

- **CONFIDENTIALITY** → preventing unauthorized access
- **INTEGRITY** → preventing modification of data
- **AVAILABILITY** → ensuring systems remain accessible

REASONS FOR CYBER SECURITY

- **Rise in cyber-crimes and hacking**
- Protection of personal and financial data
- Security of online banking and e-commerce
- **Prevention of identity theft**
- Protection against malware and ransomware
- Maintaining business continuity

CONSEQUENCES OF POOR CYBER SECURITY

- Data theft
- Financial losses
- Reputation damage
- Privacy violations

BASIC CYBER SECURITY MEASURES

- Strong passwords
- Firewalls
- Antivirus software
- Data encryption
- Regular software updates
- Multi-factor authentication

1. WHAT IS HACKING

Hacking refers to **unauthorized access or control over computer systems or networks with malicious intent**. It involves activities like finding security vulnerabilities in a system for **stealing sensitive data, deleting files, or disrupting services**.

Hacking typically involves "**cracking**" **passwords or codes that provide access to systems**, and the individual performing this act is known as a **hacker**.

❖ TYPES OF HACKING:

- **Network Hacking** : Accessing and harming networks by gathering information using tools like Telnet, Ping, NS Lookup.
- **Website Hacking**: Gaining unauthorized access to a web server or database and altering data.
- **Computer Hacking**: Breaking into a personal computer to steal info like IDs and passwords.
- **Password Hacking**: Retrieving stored passwords to bypass security.
- **Email Hacking**: Unauthorized use of an email account, for stealing info or impersonation.

❖ TYPES OF HACKER:

				
Cyber Terrorists Create fear and disruption for political/ideological reasons.	White Hat Hackers Ethical hackers improve security systems.	Black Hat Hackers Malicious hackers exploit systems for gain.	Gray Hat Hackers Blend of both Black hat & White hat Activities.	State-Sponsored Hacker Conduct cyber espionage

White Hat Hackers (Ethical Hackers)

Also known as ethical hackers, they **hack with the owner's consent to assess system security** and prevent cyberattacks. Can be employees or contractors.

Examples: *Tim Berners-Lee – Invented World Wide Web*

Working Use same methods as black hats *with permission*. Help fix issues before abuse.

Tactics/Skills: Social engineering , testing

Black Hat Hackers (Malicious Hackers)

These are malicious hackers who **disrupt systems, introduce malware, and steal confidential data**. They are also known as criminal hackers.

Examples: *Kevin Mitnick – Hacked major corps, FBI's Most Wanted*

Working Bypass security, plant malware, exploit systems for profit/disruption.

Tactics : Malware/virus creation , Phishing & identity theft , DoS attacks

Grey Hat Hackers (In Between)

A mix of white and black hat hackers, they **may hack without permission to identify vulnerabilities, often requesting payment to fix them**. Their actions are not criminal but also not fully ethical.

Examples: *Khalil Shreatch – Hacked Zuckerberg's FB to show bug*

Working :- Scan for bugs, sometimes alert or publish them without permission.

Tactics/Skills: Vulnerability scanning , Reverse engineering , Public disclosure

Red Hat Hackers (Vigilantes)

Appointed by government agencies to **identify and eliminate black hat hackers**. They are ethical but **may use illegal methods** to prevent cyberattacks.

They usually Fight black hats aggressively.. Destroy malicious systems.

Working :- Counter-hack black hats, crash their systems, expose them.

Tactics/Skills: Offensive counter-hacking , Cyber forensics , Criminal surveillance

Blue Hat Hackers (Revenge Hackers)

Hired by companies to **test new systems or software before release**, identifying vulnerabilities with the owner's consent..

Working :- Use tools or basic skills to attack ex-employers, rivals, etc.

Tactics/Skills: Ready-made exploits , Simple DoS , Email bombing/basic hacks

Green Hat Hackers (Learners)

Beginners to hacking who doesn't have malicious intent but can **unintentionally damage systems** while learning. They're eager to prove their skills, making them potentially dangerous

Working :- Use tools, tutorials, legal labs for education & practice.

Tactics/Skills: Scripting/basic exploits , Hack The Box & labs

MODULE 2 — CYBER ATTACKS

1. INTRODUCTION TO CYBER ATTACKS

A cyber-attack is an intentional attempt to gain unauthorized access to a computer system, network, or digital data for theft, disruption, spying, or destruction.

Attackers exploit vulnerabilities in systems to compromise security.

Cyber-attacks may target: Individuals , Organizations , Financial institutions , Governments , Critical infrastructure

❖ Objectives of Cyber Attacks

- Stealing confidential information
- Financial fraud
- Disrupting services
- Espionage
- Data destruction

❖ Common Sources of Attacks

- Hackers
- Cyber criminals
- Insider employees
- Terrorist organizations
- Nation-state attackers

2. CLASSIFICATION OF CYBER ATTACKS

A security attack in IT is any **attempt to compromise the confidentiality, integrity, or availability of a system or its data**. Two of most common types of attacks that can occur :-

❖ Active Attack:

An active attack occurs when a **hacker attempts to alter, manipulate, or disrupt data** or system operations. These attacks typically aim to damage or gain unauthorized access to systems.

Types of Active Attacks:

1. **Denial of Service (DoS) Attack:**
2. **Man-in-the-Middle (MitM) Attack:**
3. **Password Attacks:**

❖ Passive Attack:

A passive attack involves **observing or collecting data from a system without altering it**. These attacks are harder to detect as they don't trigger system defenses.

Types of Passive Attacks:

1. **Eavesdropping:** The attacker listens to or **intercepts communications** to steal sensitive information, such as passwords or credit card details.
2. **Port Scanning:** Attackers **scan a system to identify open ports** and services, looking for potential vulnerabilities to exploit.
3. **Traffic Analysis:** The attacker observes data flow patterns to deduce information about the exchanged data and the parties involved.

Key	Active Attack	Passive Attack
Modification	Information is modified.	Information remain unchanged.
Danger	Dangerous for Integrity.	Dangerous for Confidentiality.
Attention	Attention is to be paid on detection.	Attention is to be paid on prevention.
Impact on System	Can damage the system.	Doesn't damage any regular functioning of a system.
Victim	The victim gets informed in an active attack.	The victim does not get informed in a passive attack.
System Resources	System Resources can be changed in active attack.	System Resources are not changed in passive attack.

❖ Insider Attack:

These attacks originate from authorized users within the organization.

Examples :- Employee data theft ; Unauthorized sharing of files

❖ Outsider Attack:

Performed by external attackers without authorized access.

Examples :- Hacking attempts ; Remote malware attacks

❖ Network-Based Attacks

Target communication networks and connected systems.

Examples :- Man-in-the-Middle attack ; DoS/DDoS attack ; IP spoofing

❖ Web-Based Attacks

Attacks targeting websites and web applications.

Examples :- SQL Injection ; Cross-Site Scripting (XSS) ; Session hijacking

3. TYPES OF CYBER ATTACKS

- 1. Phishing :-** Phishing is a social engineering attack where attackers impersonate trusted organizations using emails or fake websites to steal sensitive information.

Example: Fake banking login page

- 2. Vishing :-** Vishing is voice-based phishing performed through phone calls.

Example: Fraud callers pretending to be bank officials

- 3. Snooping :-** Unauthorized observation or monitoring of data and communication.

Example: Reading confidential emails secretly

- 4. Spoofing :-** Pretending to be another user or system to gain trust.

Examples: Email spoofing ; IP spoofing

4. TYPES OF MALWARE

Malware is malicious software designed to damage, spy on, or disrupt systems.

1. ADWARE

Unwanted software that **floods your browser or screen with ads**. Often sneaks in via bundled programs or **rogue extensions** by pretending to be legitimate.

May slow down system performance and lead to more serious infections.

2. SPYWARE

Secretly **monitors our activity ,collects personal data and reports to its creator**. Runs in back, mainly logging keystrokes or screenshots . *Often installs with pop-ups or bundled software.*

Can steal sensitive data like passwords, banking info, and browsing habits.

3. VIRUS

It's a malicious program that **attaches itself to a legitimate file** or program , whenever the host file is opened , it **replicates itself** by modifying other programs too. **Needs user action** (like opening a file) to spread. *Often comes via exe files or documents.*

Requires user interaction to spread but can cause widespread file damage. ILOVEYOU Virus (2000)

4. WORM

Standalone malware that replicates itself and **spreads across networks** by exploiting vulnerabilities. They **don't need a host** or user interaction. *Commonly spreads via email or msg.*

Can quickly infect entire networks, consuming bandwidth and resources. Blaster Worm (2003)

5. TROJAN (TROJAN HORSE)

Disguises itself as useful software. Once installed, gives attackers **access to your system**, steals data, or **installs more malware** like ransomware. *Often comes via spam or fake tools.*

Grants remote access to attackers and often installs additional malware.

6. RANSOMWARE

Encrypts files or locks your device, demanding money for access. Spread via phishing or drive-by downloads. Hard to remove and often targets businesses.

Often irreversible without a decryption key; huge threat to businesses.

7. ROOTKIT

Gives attackers **root (admin) access and hides deep in the OS**, making it hard to detect or remove.

Very hard to detect and remove—may require OS reinstallation.

8. KEYLOGGER

Records everything typed on your keyboard, sending passwords, logins, financial info to attackers.

Can capture login credentials, credit card numbers, and private messages

9. EXPLOITS / ZERO-DAY EXPLOITS

Takes advantage of software bugs or security holes. Uses a **new bug or virus to steal data everytime , giving no time to make any fix** , giving full system access.

Very dangerous due to lack of available fixes or defenses.

10. BOTNET

Network of **infected systems (bots) controlled remotely by attackers** for DDoS attacks, keylogging, cryptojacking, etc.

Often used in large-scale DDoS attacks, spam campaigns, or credential theft.

5. ESSENTIAL TERMINOLOGIES :

A **potential danger** that could exploit a **vulnerability** to compromise system security, like a hacker attempting unauthorized access.

❖ TYPES OF THREATS

1. **Internal Threats** :- Originating from employees or insiders.
Examples: Data leakage ; Unauthorized access
2. **External Threats** :- Originating outside the organization.
Examples: Hackers ; Malware attacks
3. **Natural Threats** :- Caused by environmental events.
Examples: Floods ; Earthquakes ; Fire
4. **Accidental Threats** :- Unintentional actions causing damage.
Examples: Human errors ; Accidental deletion of files
5. **Intentional Threats** :- Deliberate harmful actions.
Examples: Cyber terrorism ; Data theft

❖ VULNERABILITY:

A **weakness** in a system that attackers can exploit, such as unpatched software or weak passwords.

❖ TARGET OF EVALUATION (TOE):

The system or environment **being assessed for security**, including all components like hardware, software, and configurations.

❖ ATTACK:

An **intentional action to exploit a vulnerability**, such as a **Denial of Service (DoS)** attack to disrupt services.

❖ EXPLOIT:

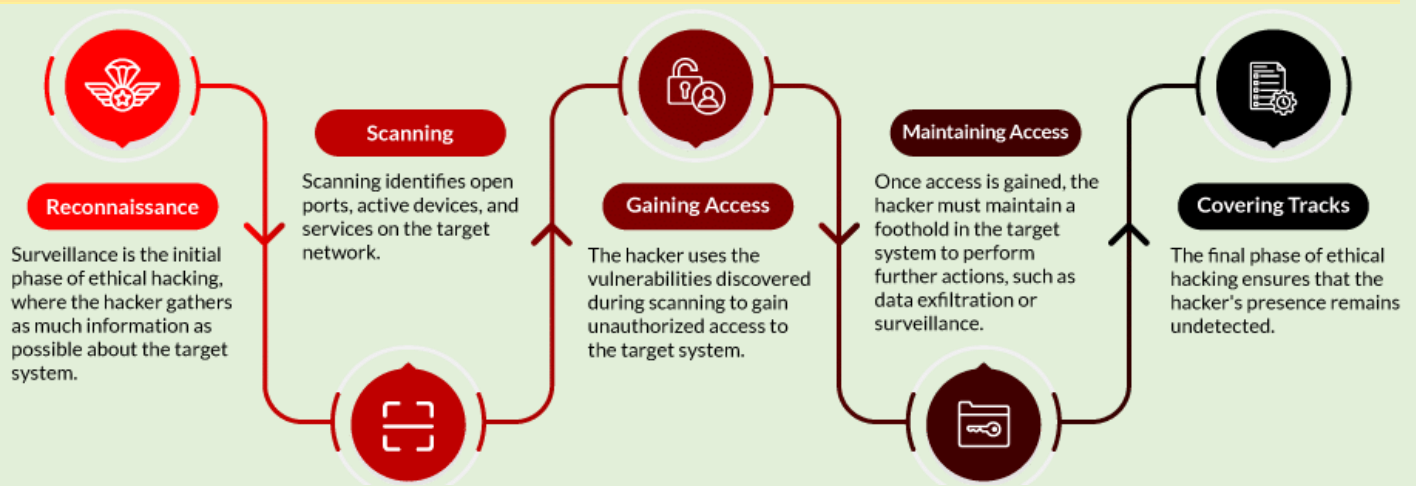
A method or **tool used to take advantage of a vulnerability**, like an **SQL Injection** to manipulate a database.

6. CONCEPT OF ETHICAL HACKING :

Hacking is usually legal as long as it is being done to **find weaknesses in a computer or network system for testing purpose**. This sort of hacking is what we call **Ethical Hacking**. Ethical Hackers help organizations to understand

- ✓ What vulnerabilities could an attacker exploit?
- ✓ What systems or data are most at risk?
- ✓ What damage could an attacker cause with the compromised information?
- ✓ How many security layers detect or log the intrusion?

7. PHASES OF ETHICAL HACKING :



1. Reconnaissance

The first phase involves **gathering information about the target system**. This step is crucial for planning further attacks.

Types:

- i. **Active Reconnaissance:** **Directly interacting with the target** system by scanning the system.
- ii. **Passive Reconnaissance:** Collecting information **without directly engaging** with the target.

2. Scanning

The scanning phase **focuses on identifying open ports, active devices**, and services within the target network, revealing areas of vulnerability.

Types:

- i. **Vulnerability Scanning:** **Identifying known system weaknesses** (e.g., using Nessus).
- ii. **Network Mapping:** **Visualizing** the network's structure (e.g., using SolarWinds).

3. Gaining Access

This phase involves **exploiting vulnerabilities to gain unauthorized access to the system**, from low-level user accounts to administrative control.

- i. **Password Cracking:** Using **brute force** or dictionary attacks.
- ii. **Exploiting Vulnerabilities:** Using **SQL Injection** or buffer overflows to gain access.

4. Maintaining Access

After gaining access, the hacker must **ensure they can re-enter the system without detection**. This involves installing backdoors or rootkits.

5. Clearing Tracks

To avoid detection, the **hacker removes evidence of the attack**, including deleting logs, modifying timestamps, and hiding files.

Techniques:

- i. **Log Tampering:** Modifying or **deleting logs** to remove traces of the attack.
- ii. **Steganography:** **Hiding malicious data** inside legitimate files.

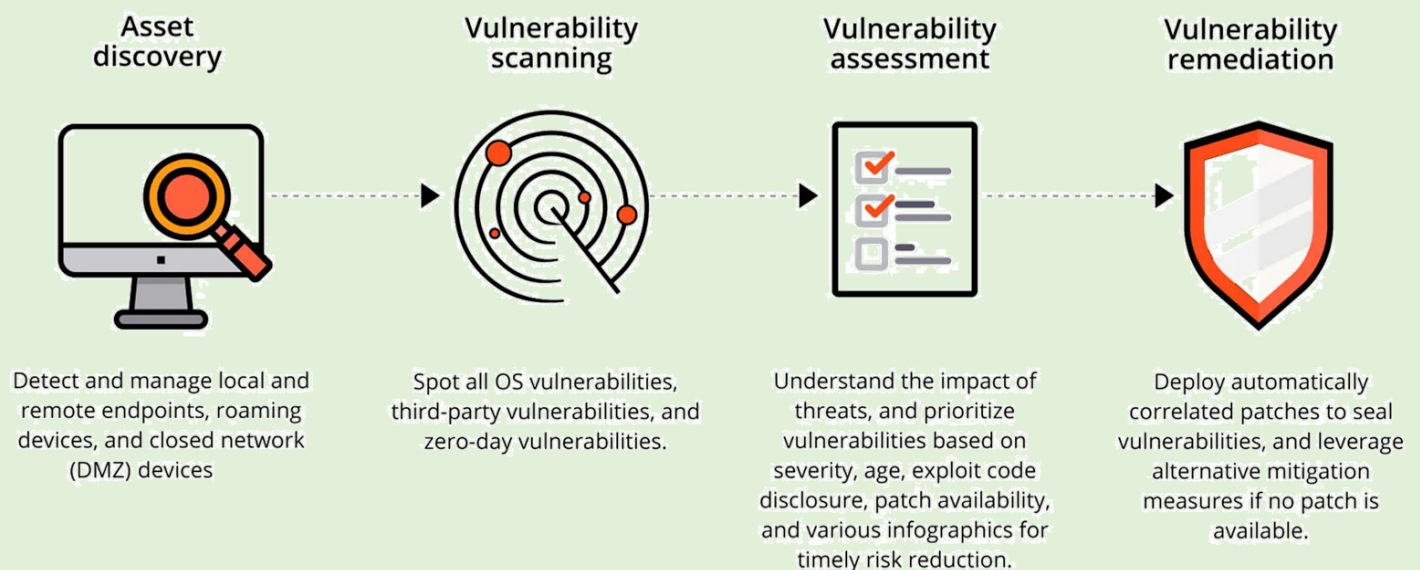
MODULE 3 — INTRUSION DETECTION AND PREVENTION

1. VULNERABILITY ASSESSMENT

Vulnerability Assessment is the **process of identifying, analyzing, and evaluating weaknesses in systems**, software, or networks that attackers may exploit. It helps organizations:

- **Detect security gaps**
- Reduce attack risks
- Improve protection mechanisms

Steps/Working in Vulnerability Assessment



Common Vulnerabilities

- Weak passwords
- Outdated software
- Open ports
- Misconfigured systems

Tools Used :- Nessus , OpenVAS , Nikto

2. INTRUSION DETECTION SYSTEM (IDS)

An Intrusion Detection System **monitors system or network activities to detect suspicious behavior**, policy violations, or attacks. IDS mainly generates alerts when threats are identified.

*IDS acts as a monitoring system but **does not actively stop attacks**.*

1. **Network-Based IDS (NIDS) :-** Monitors entire network traffic.

Example: Detecting suspicious packets

2. **Host-Based IDS (HIDS) :-** Monitors activities **within a specific computer** or device.

Example: Monitoring file modifications

Detection Methods

1. **Signature-Based Detection :-** Detects known attack patterns.
2. **Anomaly-Based Detection :-** Detects unusual activities differing from normal behavior.

Advantages

- Early threat detection
- Continuous monitoring
- Security alert generation

Limitations

- False alarms
- Cannot prevent attacks
- Limited against unknown attacks

3. INTRUSION PREVENTION SYSTEM (IPS)

It is a security system that **not only detects attacks but also automatically blocks** or prevents them in real time.

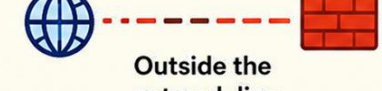
*IPS is usually placed inline within the network to **stop malicious traffic immediately**.*

FUNCTIONS OF IPS

- **Detect** malicious traffic
- **Block** suspicious packets
- **Prevent** unauthorized access
- Stop malware spread

TYPES OF IPS

1. **NETWORK-BASED IPS (NIPS)** :- Protects **entire network traffic**.
2. **HOST-BASED IPS (HIPS)** :- Protects **individual devices**.
3. **WIRELESS IPS (WIPS)** :- Protects **wireless networks** from attacks.

IDS	VS	IPS
 Intrusion Detection System	 Full name	 Intrusion Prevention System
 Monitors & detects suspicious activity	 Function	 Detects and actively blocks threats
 Alerts administrators	 Response	 Automatically takes action (e.g., blocking)
 Outside the network line	 Placement	 Inline with the traffic path
 Passive	 Action	 Active
 Snort (IDS mode)	 Example	 Snort (IPS mode), Suricata

MODULE 4 — AUTHENTICATION METHODS

1. INTRODUCTION

Authentication is the process of **verifying the identity of a user, device, or system before granting access to resources**. It ensures that only authorized users can access protected systems and information.

Authentication is the first layer of security in most systems.

OBJECTIVES OF AUTHENTICATION

- Identity verification
- Prevent unauthorized access
- Protect sensitive information

2. TYPES OF AUTHENTICATION METHODS

Authentication methods are mainly categorized into three factors.

1. KNOWLEDGE-BASED AUTHENTICATION

Based on **something the user knows**. If credentials match, access is granted; otherwise denied .

Examples :- Passwords , PINs , Security questions

Advantages

- Easy to implement
- Low cost

Limitations

- Weak passwords can be guessed
- Vulnerable to phishing

2. POSSESSION-BASED AUTHENTICATION

Based on **something physical or digital the user possesses**.

Examples:- Smart cards , OTP tokens , Mobile authentication apps

Advantages

- Stronger than passwords alone

Limitations

- Device can be stolen or lost

3. BIOMETRIC AUTHENTICATION

Based on **physical or behavioral characteristics**. Some examples include - Fingerprint recognition , Face recognition , Iris scanning , Voice recognition .

Advantages

- Difficult to duplicate
- Convenient for users

Limitations

- Expensive hardware
- Privacy concerns

3. AUTHENTICATION TECHNIQUES

1. SINGLE-FACTOR AUTHENTICATION (SFA)

Uses one verification factor, usually a username and password. Access is granted when credentials match stored values. Common in systems **prioritizing ease of access over high security**; security **depends mainly on password strength**.

ADVANTAGES:

- **Simple and fast**
- Easy to manage
- Low cost

2. TWO-FACTOR AUTHENTICATION (2FA)

Uses two authentication factors, usually **something the user knows (password) and something the user has (OTP/token)**. The second factor adds extra protection even if one factor is compromised.

ADVANTAGES:

- **Better security** than SFA
- Reduces password-related risks

3. MULTI-FACTOR AUTHENTICATION (MFA)

Uses two or more independent factors such as **passwords, tokens, and biometrics** for identity verification. Mainly **used in high-security environments** to protect sensitive data.

ADVANTAGES:

- Strong access protection
- Reduces credential compromise risks

4. BIOMETRIC AUTHENTICATION METHODS

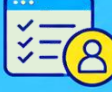
Biometric authentication identifies users through unique biological or behavioral traits.

❖ **PHYSIOLOGICAL BIOMETRICS :-** Based on body characteristics.

Examples: Fingerprint , Face , Iris , Retina , Palm print

❖ **BEHAVIORAL BIOMETRICS :-** Based on user behavior patterns.

Examples: Voice recognition , Signature analysis , Typing rhythm

Authentication		vs	Authorization	
	Verifying a user's identity before giving them permission to access a system, account, or file.			Verifying a user's access level to a system, account, or file.
	To confirm the user's identity and prevent unauthorized access			To ensure users can only access resources they are allowed to
	Compares user credentials with stored data			Grants or denies access based on roles/permissions
	Username/Password, OTP, Security Questions			Role-Based Access Control (RBAC), Permissions
		Definition		
		Purpose		
		Process		
		Methods		

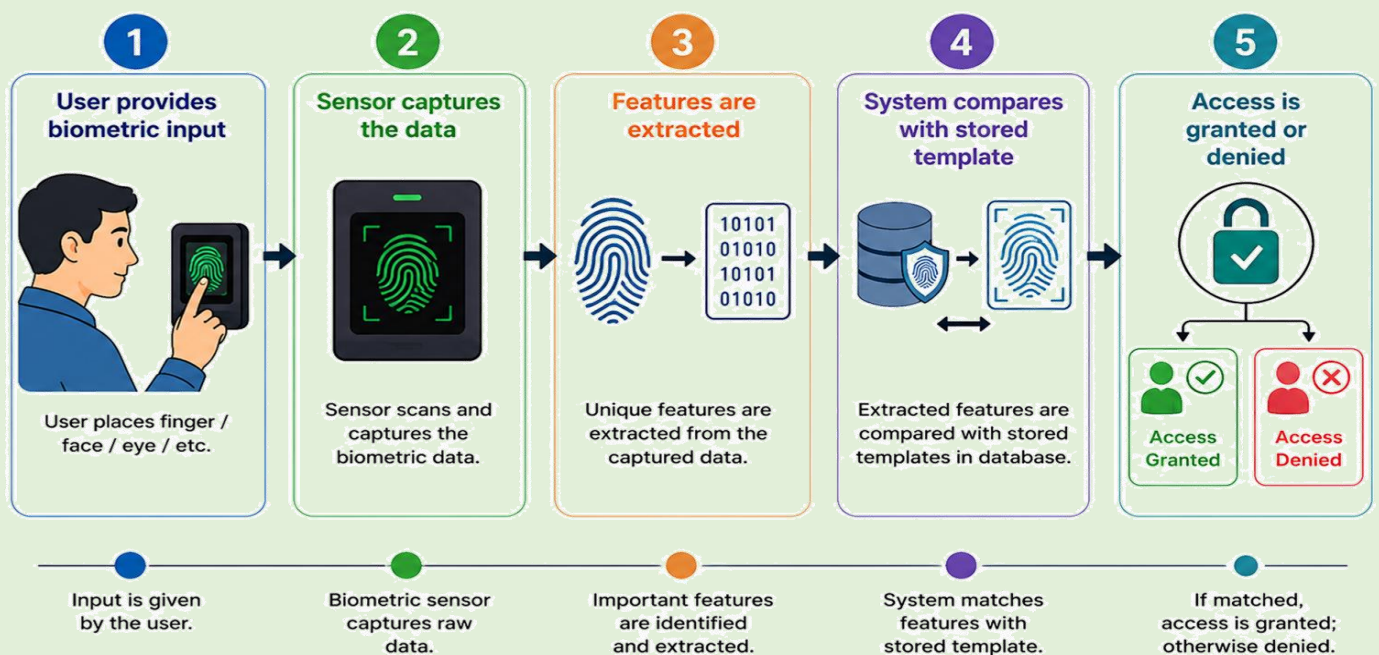
5. BIOMETRIC SYSTEMS

A biometric system captures, stores, and compares biometric characteristics for identification or verification.

COMPONENTS OF BIOMETRIC SYSTEM

1. **SENSOR** :- Captures biometric input. **Example:** Fingerprint scanner
2. **FEATURE EXTRACTION MODULE** :- Extracts unique features from the biometric sample.
3. **DATABASE** :- Stores biometric templates.
4. **MATCHING MODULE**:- Compares input data with stored records.
5. **DECISION MODULE** :- Determines whether authentication is successful.

WORKING OF BIOMETRIC SYSTEM



ADVANTAGES OF BIOMETRIC SYSTEMS

- **High security**
- Difficult to forge
- **Fast authentication**
- User convenience

LIMITATIONS OF BIOMETRIC SYSTEMS

- **Expensive** implementation
- **Privacy concerns**
- Errors due to poor sensor quality
- Cannot be changed easily if compromised

MODULE 5 – SECURITY MODELS

1. Different Security Models and Security Mechanisms

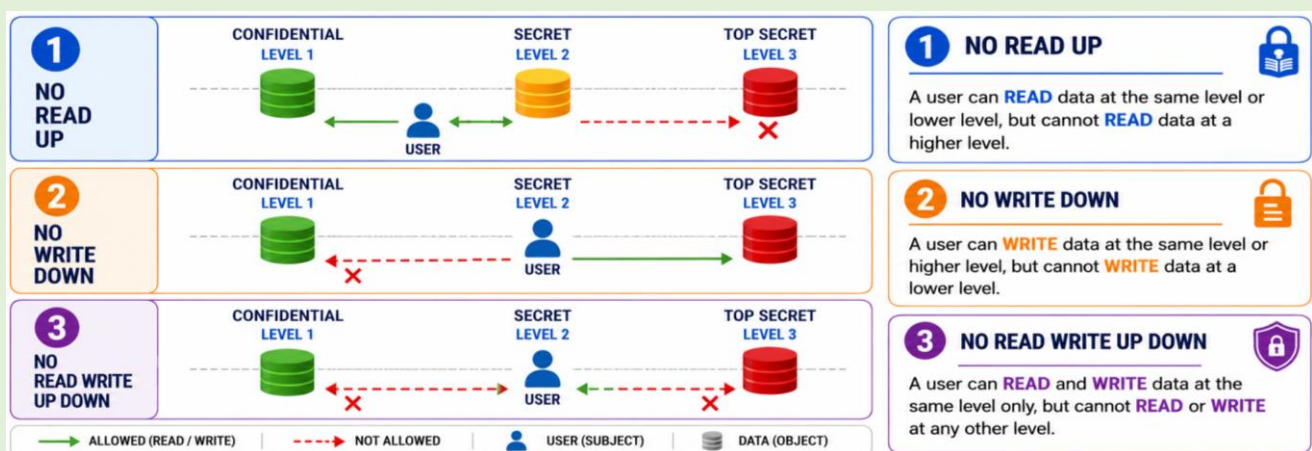
1. CONFIDENTIALITY-FOCUSED MODELS

Bell-LaPadula (BLP) Model

A formal security model that protects **data confidentiality** by preventing unauthorized disclosure of sensitive information. It is mainly used in military and government environments.

*Information should flow only from **lower to higher security levels**, preventing confidential data from leaking to lower classifications.*

KEY RULES:



CHARACTERISTICS:

- Uses security classifications (Confidential, Secret, Top Secret).
- Access is based on security labels.
- Commonly implemented through **Mandatory Access Control (MAC)**.

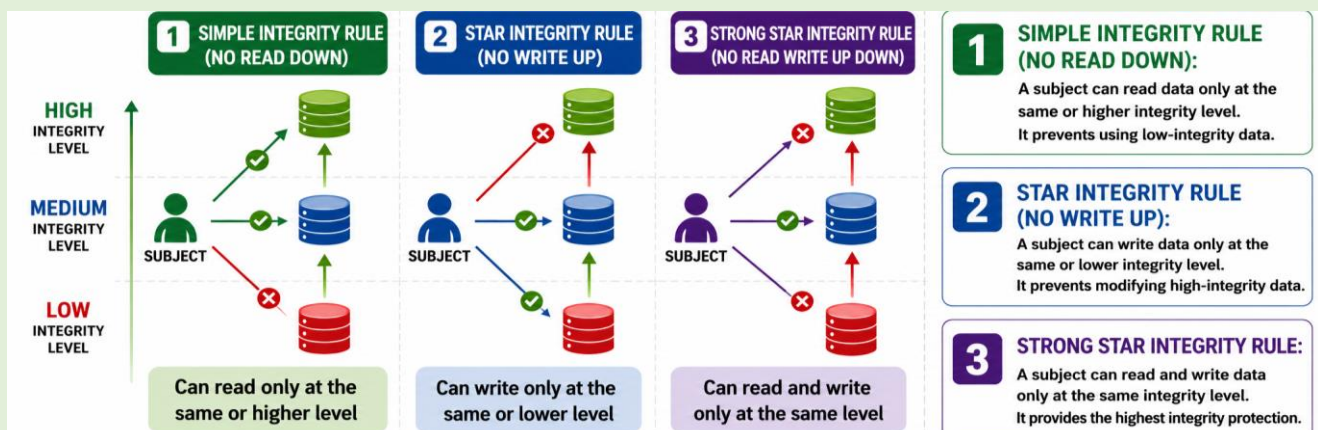
2. INTEGRITY-FOCUSED MODELS

Biba Model

A formal model that protects **data integrity** by preventing unauthorized or incorrect modifications.

*Information should flow only from **higher to lower integrity levels**, preventing trusted data from being affected by untrusted sources.*

KEY RULES:



CHARACTERISTICS:

- Inverse of Bell-LaPadula.
- Prioritizes integrity over confidentiality.
- Suitable where data accuracy is critical.

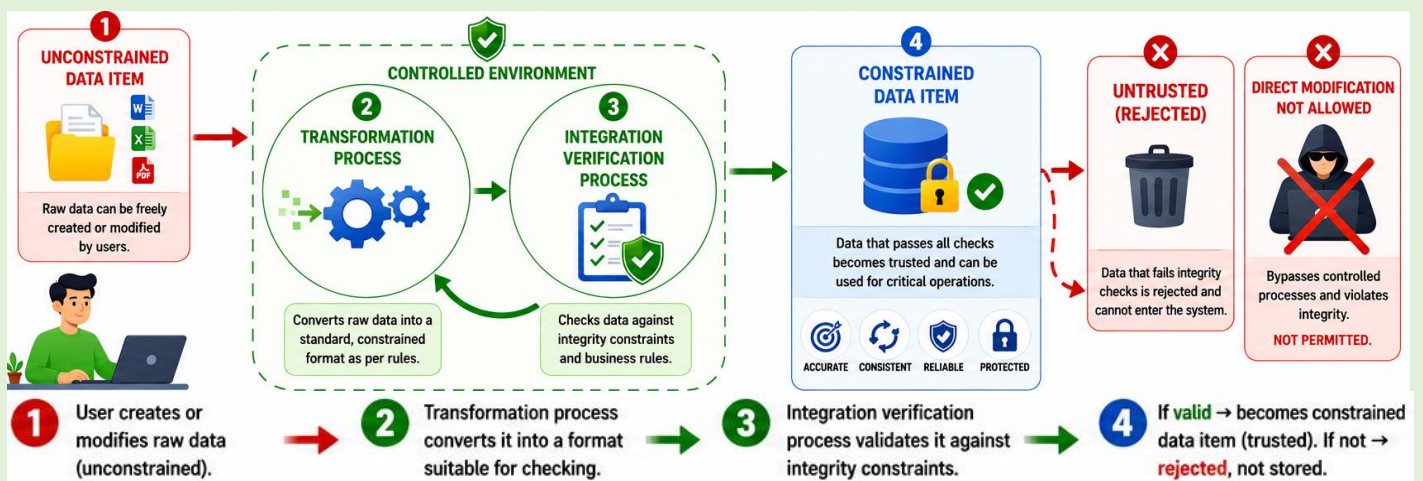
Clark-Wilson Model

An integrity model designed for commercial and financial systems to maintain accurate and consistent data.

*Data can only be modified through **authorized transformation procedures**, with responsibilities divided among users.*

Components:

1. **Subject:** The user requesting access.
2. **Constrained Data Items (CDIs):** Critical data that cannot be accessed or modified directly.
3. **Unconstrained Data Items (UDIs):** Non-critical data that can be accessed directly.
4. **Transformation Procedure (TP):** Authorized program that validates and performs modifications on CDIs.
5. **Integrity Verification Procedure (IVP):** Verifies data integrity through authentication, authorization, and consistency checks before allowing access.

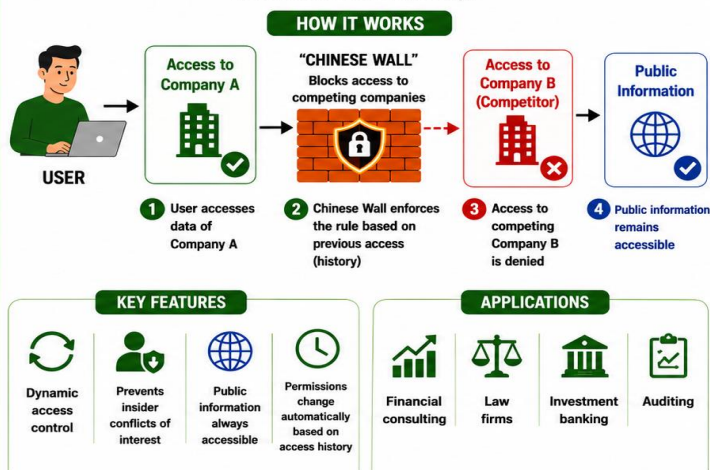


Applications: Banking ; Financial systems ; Enterprise accounting

3. HYBRID AND CONFLICT-OF-INTEREST MODELS

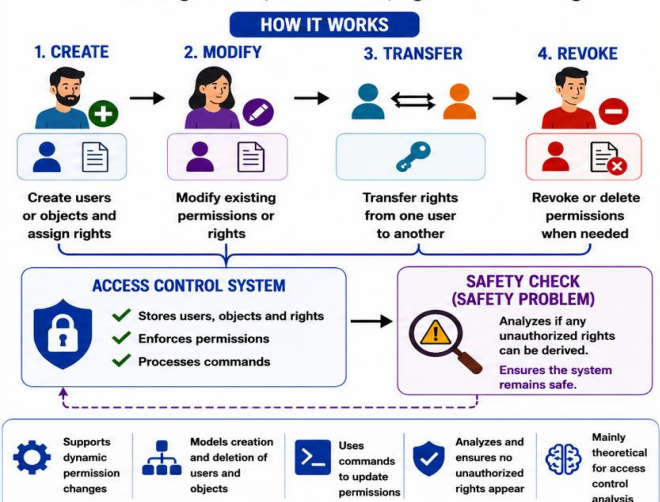
BREWER-NASH (CHINESE WALL) MODEL

Prevents conflicts of interest by blocking access to competing organizations based on a user's access history.



HARRISON-RUZZO-ULLMAN (HRU) MODEL

A formal model for managing and analyzing access rights and ensuring unsafe (unauthorized) rights cannot emerge.



4. ACCESS CONTROL MODELS

Mandatory Access Control (MAC)

A centralized model where the system enforces access using predefined security labels and clearance levels.

Only the system defines permissions; users cannot modify access rights.

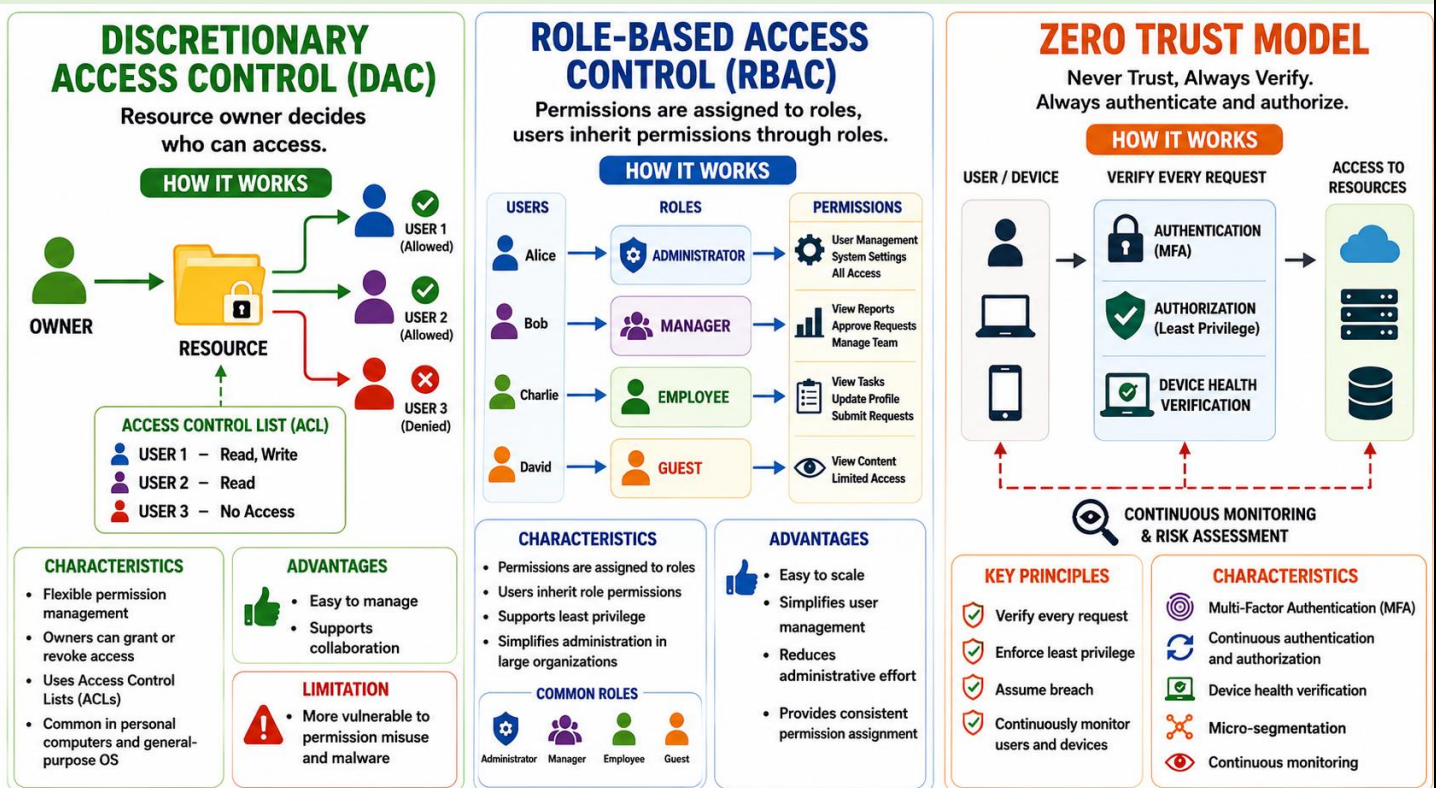
CHARACTERISTICS:

- Highly secure and rigid.
- Based on security labels and clearances.
- Users cannot grant permissions.
- Common in military and government systems.



Advantages: Strong confidentiality and Prevents unauthorized permission changes.

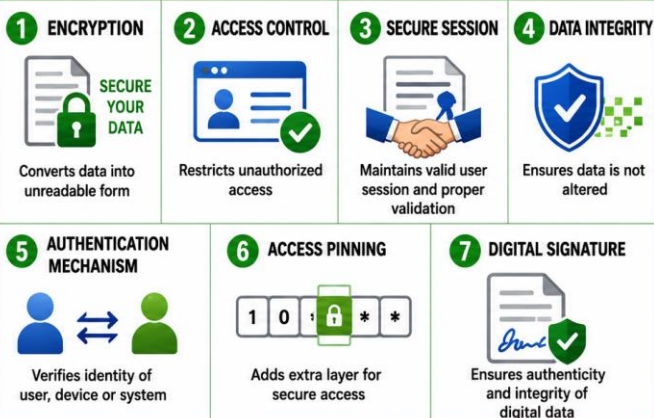
Limitation: Less flexible and harder to administer.



4. SECURITY MECHANISMS

1. SPECIFIC SECURITY MECHANISMS

Logical / Legal / OS / App & Dynamic (Digital) Security Controls



2. PERVASIVE SECURITY MECHANISMS

Apply Across All Ends, to All Devices/Connections & Critical Information



Specific Security Mechanisms	Pervasive Security Mechanisms
Applied to specific OSI layers	Applied across all OSI layers
Provide encryption , authentication , etc	Support overall security and policy enforcement
Focus on protecting particular data	Ensure security management and system-wide controls
Examples: Access Control, Digital Signature	Examples: Trusted Functionality, Security Labels,
Technical controls at specific layers	Management and infrastructural controls
Protects individual communications or data	Maintains security of entire system

2. Information Security and Network Security

Information security refers to the **protection of computer systems**, networks, software, and data from cyber attacks, unauthorized access, theft, and damage.

Confidentiality **I**ntegrity **A**vailability

CONFIDENTIALITY → preventing unauthorized access

INTEGRITY → preventing modification of data

AVAILABILITY → ensuring systems remain accessible

3. NETWORK SECURITY

Network Security protects networks and transmitted data from **unauthorized access, misuse, and cyberattacks**, ensuring secure and reliable communication.

- Prevents unauthorized access and cyber threats.
- Ensures **Confidentiality, Integrity, and Availability (CIA)**.

WORKING OF NETWORK SECURITY :-

Network security uses **multiple layers of protection** to control access and block threats before they reach network resources.

1. PHYSICAL NETWORK SECURITY:

Protects network hardware from physical attacks and unauthorized access.

- Uses locks, biometrics, CCTV, and access cards.
- Secures servers, routers, switches, and data centers.

2. TECHNICAL NETWORK SECURITY

Protects data using software and security technologies.

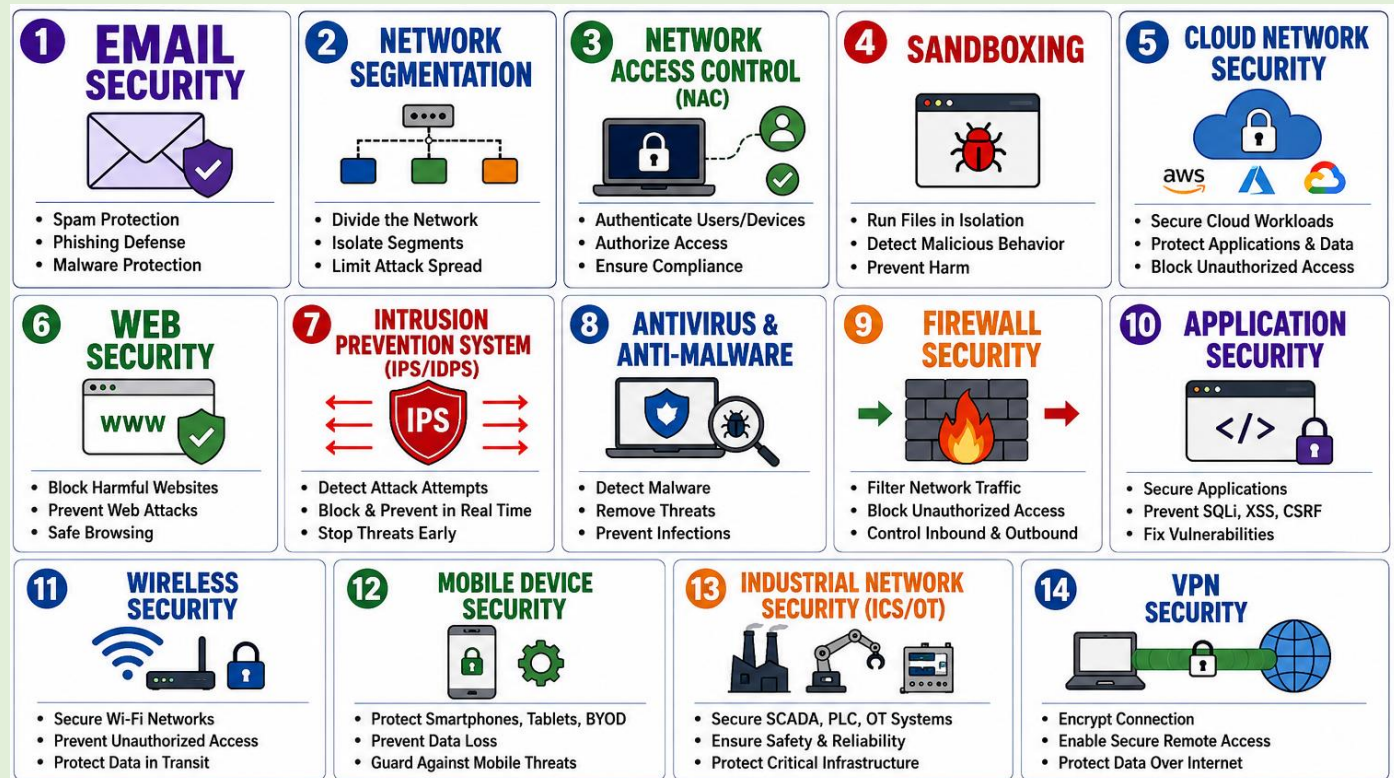
- Secures data in transit and at rest.
- Uses firewalls, encryption, antivirus, and IDS/IPS.

3. ADMINISTRATIVE NETWORK SECURITY:

Uses policies and procedures to manage secure network access.

- Defines authentication, authorization, and user roles.
- Enforces security policies, monitoring, and compliance.

TYPES OF NETWORK SECURITY :-



4. Operating System Security

Operating System (OS) Security is the **set of mechanisms that protect the operating system**, its resources, and user data from unauthorized access, misuse, modification, and cyber attacks.

*Ensure only **authorized users and processes** can access system resources while maintaining **Confidentiality, Integrity, and Availability (CIA)**.*

Confidentiality **I**ntegrity **A**vailability

CONFIDENTIALITY → preventing unauthorized access

INTEGRITY → preventing modification of data

AVAILABILITY → ensuring systems remain

Common Threats :-

1. Malware

Malicious software designed to damage systems, steal data, or gain unauthorized access.

✓ **Examples:** Virus, Worm, Trojan Horse, Ransomware, Spyware.

2. Buffer Overflow

Occurs when a program writes more data than a buffer can hold, overwriting adjacent memory and allowing execution of malicious code.

✓ **Working:** Oversized input → Buffer overflow → Return address overwritten → Malicious code executes.

3. Network Intrusion

Unauthorized access through a network by exploiting vulnerabilities.

✓ **Detection:** Intrusion Detection System (IDS).

Types of Threats :-

A. PROGRAM THREATS

- ✓ **Virus:** Attaches to executable files and spreads when executed.
- ✓ **Trojan Horse:** Malicious software disguised as legitimate software.
- ✓ **Logic Bomb:** Malicious code activated only when a specific trigger occurs.

B. SYSTEM THREATS

- ✓ **Worm:** Self-replicates across networks without user interaction.
- ✓ **Denial of Service (DoS):** Exhausts system resources, making services unavailable.

C. SECURITY MECHANISMS

1. **Authentication:** Verifies user identity (Passwords, Biometrics, MFA).
2. **Authorization:** Determines what authenticated users are allowed to access.
3. **Access Control:** Restricts access using permissions (Read, Write, Execute, ACLs).
4. **Encryption :** Protects stored and transmitted data from unauthorized access.

D. COMMON VULNERABILITIES

- Weak or stolen passwords
- Trap doors (Backdoors)
- Invalid input parameters
- Improper access control
- Line tapping
- Electronic data capture
- Lost communication sessions
- Waste recovery
- Rogue software

Basic Security Measures

- Strong passwords and MFA
- Regular updates and patching
- Antivirus/Anti-malware
- Firewalls
- File permissions and access control
- Encryption
- IDS/IPS monitoring
- Regular backups

MODULE 6 – ONLINE SECURITY

1. WEB SECURITY

Web security is a **branch of cybersecurity focused on protecting websites, web applications, and APIs from internet-based attacks.**

KEY THREAT VECTORS:-

Malicious actors exploit various vulnerabilities to compromise web assets :-

1. SQL Injection (SQLi):

Attackers **insert malicious SQL code into input fields to manipulate backend databases** and access restricted data.

2. Cross-Site Scripting (XSS):-

Hackers **inject malicious scripts into trusted websites**, which then execute in the victims' browsers, often stealing session data or credentials

3. DDoS Attacks:

Distributed Denial of Service (DDoS) attacks flood a web server with traffic, causing it to crash or become unavailable to legitimate users

4. Phishing & Malware:

Redirecting users or tricking them into downloading malicious files, reveal personal information

CORE SECURITY MECHANISMS :-

- ❖ **Web Application Firewalls (WAF):** Filters and monitors incoming HTTP traffic to block malicious requests and prevent exploits like SQLi and XSS.
- ❖ **Encryption (SSL/TLS):** Secures data in transit between the user's browser and the web server, ensuring it cannot be intercepted and read.
- ❖ **Authentication and Access Control:** Verifies user identity using strong passwords, Multi-Factor Authentication (MFA), and role-based permissions.

WEB SECURITY BEST PRACTICES



USE HTTPS
ALWAYS USE HTTPS TO ENCRYPT DATA IN TRANSIT.



VALIDATE INPUT
ALWAYS VALIDATE AND SANITIZE USER INPUT TO PREVENT ATTACKS.



USE SECURE CODE
FOLLOW SECURE CODING PRACTICES AND AVOID COMMON VULNERABILITIES.



STRONG AUTHENTICATION
USE STRONG PASSWORDS AND MULTI-FACTOR AUTHENTICATION (MFA).



KEEP UPDATED
REGULARLY UPDATE YOUR SOFTWARE, PLUGINS AND FRAMEWORKS.



USE FIREWALL & IDS
USE WEB APPLICATION FIREWALL (WAF) AND INTRUSION DETECTION SYSTEMS.

TOP WEB SECURITY TOOLS FOR DEVELOPERS & SECURITY EXPERTS



OWASP ZAP

A FREE, OPEN-SOURCE TOOL USED TO FIND SECURITY FLAWS IN WEB APPLICATIONS.



BURP SUITE

A POWERFUL PLATFORM FOR ANALYZING AND TESTING WEB SECURITY.



CLOUDFLARE

OFFERS PROTECTION AGAINST DDOS ATTACKS AND ENHANCES PERFORMANCE WITH CDN SERVICES.



SSL LABS

EVALUATES THE SECURITY OF YOUR SSL/TLS SETUP AND PROVIDES A DETAILED REPORT.



SNYK

DETECTS VULNERABILITIES IN OPEN-SOURCE LIBRARIES AND HELPS FIX THEM QUICKLY.

2. EMAIL SECURITY

Email is a daily communication tool heavily targeted by hackers for phishing, malware, and password theft. Protection requires combining technical tools with smart habits.

TECHNICAL TOOLS

- **Encryption:** Keeps message content private.
- **Filters:** Blocks junk mail automatically.

SMART HABITS

- **Spotting fakes:** Identifying phishing scams.
- **Passwords:** Preventing unauthorized access.

WAYS TO PROTECT YOUR EMAIL	TYPES OF EMAIL THREATS	EMAIL SECURITY BEST PRACTICES	KEY EMAIL SECURITY OBJECTIVES
 01 Use a Strong, Unique Password	 01 Phishing	 Stop phishing attacks	 Zero-Day Protection
 02 Avoid Public Wi-Fi for Email	 02 Email Bombing	 Encrypt data	 Real-Time Threat Detection
 03 Enable Two-Factor Authentication (2FA)	 03 Spam	 Control spam	 Communication Confidentiality
 04 Check Login Activity Regularly	 04 Email Spoofing	 Prevent data loss	 Ransomware Protection
 05 Beware of Phishing Scams	 05 Social Engineering	 Protect from malware	 Device Access Control
	 06 Malware / Virus	 Protect confidential information	 Spam & Phishing Protection

TOP EMAIL SECURITY TOOLS					
 Microsoft Defender for 365	 Google Workspace Security	 Proofpoint Email Protection	 Mimecast Email Security	 Trend Micro Email Security	 Barracuda Email Protection
Advanced threat protection for business email.	Filters spam, phishing & malware in your inbox.	Protects against advanced threats, phishing & malware.	Cloud security for email threats, continuity & resilience.	Blocks threats, spam and ransomware in emails.	Comprehensive email threat protection & archiving.

3. MOBILE DEVICE SECURITY

Mobile Device Security is the process of protecting smartphones, tablets, and other portable devices from unauthorized access, malware, data theft, and cyber-attacks. It ensures the **Confidentiality, Integrity, and Availability (CIA)** of data stored on or transmitted by mobile devices.

MOBILE SECURITY BEST PRACTICES					
 1 LOCK SCREEN SECURITY Use PIN, password, fingerprint, or face unlock to prevent access.	 2 ENCRYPTION Encrypts data so it cannot be read without proper authentication.	 3 APP PERMISSIONS Allow only necessary permissions (Camera, Microphone, Location). Avoid unnecessary access.	 4 SOFTWARE UPDATES Regularly update OS and apps to fix security vulnerabilities.	 5 SECURE NETWORK USAGE Avoid public Wi-Fi for sensitive activities. Use a VPN on public networks.	 6 BACKUP Regularly back up important data to cloud or external devices to prevent data loss.

COMMON MOBILE THREATS				
 a MALWARE Malicious software that steals data or damages your device.	 b PHISHING Fake emails, SMS, or websites trick you into revealing sensitive info.	 c UNSECURED PUBLIC WI-FI Attackers can intercept your data on open Wi-Fi networks.	 d APP VULNERABILITIES Flaws in apps can be exploited to steal data or install malware.	 e LOST OR STOLEN DEVICES Without proper protection, your data can be accessed by others.

4. CLOUD SECURITY

Cloud security refers to the technologies, policies, and practices designed to **protect cloud-based applications, data, and infrastructure**. It relies on the SRM.

SHARED RESPONSIBILITY MODEL :- Cloud providers secure the physical infrastructure and Clients are responsible for securing their data, access controls, and configurations

❖ 5 CORE COMPONENTS :-

1. IDENTITY AND ACCESS MANAGEMENT (IAM)

Controls **who access cloud resources & permitted actions**. It gives users only needed permissions.

Mechanism: **Uses Multi-Factor Authentication** and Role-Based Access Control (RBAC).

Purpose: Prevent unauthorized access.

2. DATA ENCRYPTION

Protects sensitive information by **converting it into unreadable ciphertext so that data remains unreadable if intercepted** without the decryption key.

Purpose: Protect data confidentiality.

3. CLOUD SECURITY POSTURE MANAGEMENT (CSPM)

Continuously scans cloud resources to identify misconfigurations, security weaknesses, and compliance violations and alerts issues prior to any mishap.

Example :- Publicly exposed storage buckets, open database ports, and excessive permissions.

Purpose: Detect configuration risks early.

4. ZERO TRUST ARCHITECTURE

Never Trust, Always Verify.

Any user, device, or application is **not automatically trusted inside or outside the network**. **Every request must be authenticated**, authorized, and continuously verified.

Purpose: Minimize unauthorized access.

5. THREAT DETECTION AND RESPONSE

After Identifying suspicious activities it **uses automated or manual response mechanisms to contain and recover** from attacks.

Purpose: Detect, respond to, and recover from security incidents.

❖ FOUR PILLARS :-



MODULE 7 – IoT & Social Media Security

1. IoT SECURITY

IoT (Internet of Things) is a **network of physical devices connected to the internet** that can collect, share, and exchange data automatically.

IoT security refers to protecting these internet-connected devices (sensors, smart appliances, wearables) and their networks from cyber threats.

Example :- **Mirai Botnet (2016)** – compromised thousands of IoT devices (cameras, routers) to launch massive DDoS attack on Dyn DNS, taking down Twitter, Netflix, etc.

SECURITY WEAKNESSES IN IOT



- 1. Limited Processing Power:** Many IoT devices cannot support strong encryption and advanced security features.
- 2. Weak Authentication:** Default or weak passwords allow unauthorized access and remote control of devices.
- 3. Lack of Firmware Updates:** Outdated software contains known vulnerabilities that attackers can exploit.
- 4. Insecure Communication:** Unencrypted data transmission can be intercepted, causing data leakage.
- 5. Large Attack Surface & Poor Security Standards:** Billions of connected devices and weak manufacturer security practices make IoT networks easier to attack.

2. CYBER PHYSICAL SYSTEM [CPS]

A cyber-physical system is a **combination of closely integrated physical processes, networking, and computation**. The physical process is monitored and **controlled by embedded (cyber) subsystems via networked systems with a feedback loop** to change their behaviour when needed.

Physical systems + digital systems = Cyber Physical System.

Example :- In a smart AC, the temperature sensor (physical device) sends data to the controller (computing component), which turns the AC (physical device) on or off based on the temperature. Updated temperature is measured again, forming a **feedback loop**.

Smart Grid ; Autonomous (Self-driving) Cars ; Smart Traffic Lights ; Smart Healthcare Equipment ; etc

CPS SECURITY RISKS

1. PHYSICAL EQUIPMENT DAMAGE

Attacks can effect industrial machines, **causing equipment failure** or disruption of critical services.

- ✓ **Security Measures:** Strong authentication, network monitoring, regular software updates.
- **Example:** Attackers modify power plant controls, causing machines to overheat.

2. SENSOR DATA MANIPULATION

Attackers **alter sensor readings**, causing systems to make incorrect or unsafe decisions.

- ✓ **Security Measures:** Encrypt sensor communication, intrusion detection, data validation.
- **Example:** A temperature sensor reports **20°C** instead of **80°C**, preventing cooling.

3. UNAUTHORIZED REMOTE ACCESS

Weak authentication allows **attackers to remotely control connected devices** and systems.

- ✓ **Security Measures:** Strong authentication, authorization, secure remote access.
- **Example:** A hacker remotely unlocks a smart door or controls factory machinery.

4. SYSTEM FAILURE & SAFETY RISKS

Cyber attacks or system failures can **disrupt operations and endanger people** or property.

- ✓ **Security Measures:** Regular updates, backup & disaster recovery, continuous monitoring.
- **Example:** Incorrect commands cause a self-driving car to crash.

5. DATA THEFT

Sensitive information stored or transmitted by connected devices may be stolen by attackers.

- ✓ **Security Measures:** Encrypt communication, access control, regular security updates.
- **Example:** Compromised medical devices leak patient records.

3. IOT SECURITY REQUIREMENTS [CPS]



4. WHAT IS SOCIAL MEDIA SECURITY?

Social Media Security involves protecting personal/organizational accounts, data, and reputation on social media platforms from hacking, impersonation, and misuse.



MODULE 8 – SECURITY & VIRTUAL CURRENCY

Virtual currency is a **digital representation of value that can be digitally traded** and functions as a **medium of exchange**, a **unit of account**, and/or a **store of value**.

Unlike traditional currency, **most digital currencies are decentralized**, meaning **transactions are verified by a blockchain network instead of banks** or governments.

Blockchain = A decentralized ledger that securely records and verifies all transactions.

It provides benefits such as accessibility, speed, and reduced transaction costs, yet poses challenges including regulatory issues, cybersecurity threats, and potential for illegal use.

1. TYPES OF DIGITAL CURRENCY :-

1. CRYPTOCURRENCIES

Cryptocurrencies are **decentralized blockchain-based digital currencies** secured by cryptography, enabling **peer-to-peer transactions** without banks or governments.

- **Decentralized & Secure:** **Blockchain + cryptography** ensure secure, private transactions.
- **Limited Supply:** **Fixed supply creates scarcity** and potential value.

Examples: Bitcoin (BTC), Ethereum (ETH)

2. CENTRAL BANK DIGITAL CURRENCY (CBDC)

CBDCs are **digital versions of a country's fiat currency**, issued and regulated by the **central bank** with legal tender status.

- **Government-Backed:** Official currency **recognized by law**.
- **Centralized:** Fully **controlled and integrated with banking** systems.

Examples: Digital Yuan (e-CNY), e-Krona

3. STABLECOINS

Stablecoins are **blockchain-based digital currencies** whose value is **pegged to reserve assets** to minimize price volatility.

- **Stable Value:** Backed by fiat, commodities, or cryptocurrencies.
- **Fast & Transparent:** **Quick transactions** with reserve-backed transparency.

Examples: Tether (USDT), USD Coin (USDC), Dai

4. DIGITAL TOKENS

Digital Tokens are **blockchain-based digital assets representing ownership, rights, or services**, powered by smart contracts.

- **Asset Representation:** **Represents assets**, ownership, voting, or services.

Examples: Utility Tokens, Security Tokens

5. ELECTRONIC MONEY (E-MONEY)

E-Money is **digitally stored fiat currency managed by banks** or **payment providers** for cashless electronic payments.

- **Centralized:** Wallet or bank-based instant digital payments.

Examples: PayPal, Venmo

2. TYPES OF VIRTUAL CURRENCY :-

A. BASED ON CONVERTIBILITY

1. CLOSED (NON-CONVERTIBLE) VIRTUAL CURRENCY

Closed virtual currencies are **used only within a specific application or network** and generally cannot be exchanged back into other forms of money.

- **Restricted Use:** **Valid only within their own ecosystem.**
- **Non-Convertible:** Usually cannot be converted into fiat or other currencies.
 - **Examples:** Game Coins, PUBG UC, Roblox Robux, Loyalty Points.

2. OPEN (CONVERTIBLE) VIRTUAL CURRENCY

Open virtual currencies have **value beyond their original platform and can be exchanged for other virtual currencies or fiat currency.**

- **Convertible:** **Exchangeable for virtual currencies** and, in some cases, fiat currency.
- **External Value:** Accepted beyond their original platform.
 - **Examples:** Bitcoin (BTC), Ethereum (ETH), Tether (USDT).

B. BASED ON CONTROL (GOVERNANCE)

1. CENTRALIZED VIRTUAL CURRENCY

Centralized virtual currencies are **issued and managed by a single authority** responsible for all transactions.

- **Single Authority:** **Controls issuance, verification**, and transaction records.
 - **Examples:** Centralized in-game currencies, company-issued digital currencies.

2. DECENTRALIZED VIRTUAL CURRENCY

Decentralized virtual currencies **operate without a central authority and rely on network consensus** to validate transactions.

- **Consensus-Based:** Transactions are **verified by the network.**
- **Irreversible:** Confirmed transactions are **difficult to reverse.**
 - **Examples:** Bitcoin (BTC), Ethereum (ETH).

ADVANTAGES OF CRYPTOCURRENCIES



DECENTRALIZATION

Operate independently of governments and central banks, giving users more control over their finances.



LOWER TRANSACTION FEES

Transactions with cryptocurrencies typically have lower fees compared to traditional banking and payment systems.



GLOBAL ACCESSIBILITY

Cryptocurrencies enable borderless transactions, promoting financial inclusion and allowing people to send and receive money worldwide.

DISADVANTAGES OF CRYPTOCURRENCIES



HIGH VOLATILITY

Cryptocurrency prices can be highly volatile, leading to uncertainty and significant risk for investors.



LACK OF REGULATION

Regulatory uncertainty can create challenges for cryptocurrency adoption and legitimacy in many jurisdictions.



SECURITY CONCERNS

Cryptocurrency wallets and exchanges are vulnerable to hacking and fraud, putting users' funds at risk.

3. BLOCKCHAIN:-

Blockchain is a **decentralized, distributed digital ledger** that securely records transactions across a network of computers (nodes) without a central authority.

CORE CONCEPTS

1. DECENTRALIZATION

Data is stored across a **peer-to-peer network** (*computers connected directly to each other*) instead of a single central server, **eliminating a single point of failure**.

3. CONSENSUS MECHANISMS

Consensus Mechanisms are **protocols** (*rules used by the network*) that allow all nodes to agree **whether a transaction is valid**.

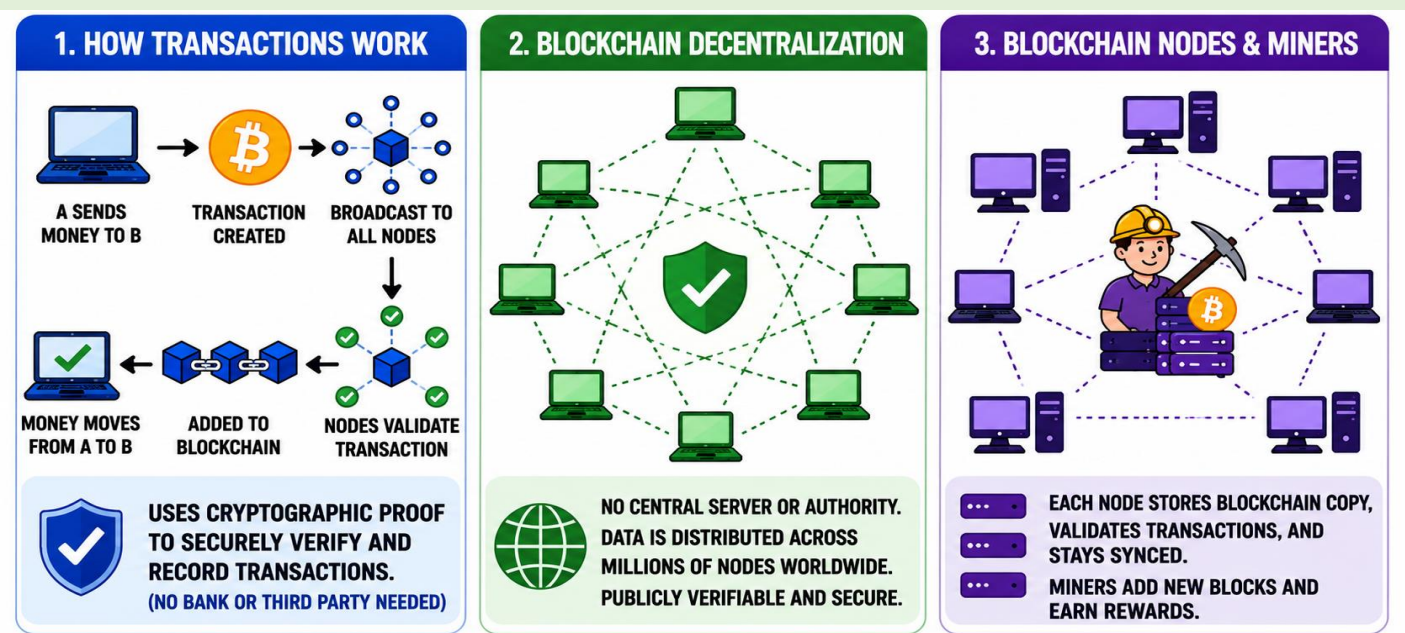
-: Examples :-

- **Proof of Work (PoW)** → Computers solve complex mathematical problems.
- **Proof of Stake (PoS)** → Validators are selected based on the amount of cryptocurrency they hold.

3. SMART CONTRACTS

Smart Contracts are **self-executing digital contracts** that automatically perform an action when **predefined conditions** (*rules written in the contract*) are satisfied.

WORKING



1. DATA BLOCKS :- Transactions and timestamps are grouped into chronological blocks.

2. CRYPTOGRAPHIC HASHING :- Each block gets a **unique digital fingerprint (hash)** based on its own data and the previous block's hash, securely linking all blocks into a chain.

3. NETWORK CONSENSUS :- Every node maintains an identical copy of the ledger, and the majority must verify new transactions before they are added.

SECURITY FEATURES

1. IMMUTABILITY

Once a block is verified and added to the blockchain, it **cannot be modified or deleted** unless the network agrees (*network consensus*), making records permanent and tamper-resistant.

2. TAMPER RESISTANCE

Changing any confirmed block breaks the cryptographic chain, causing unauthorized modifications to be detected and rejected by the network.

TYPES OF BLOCKCHAIN

1. PUBLIC BLOCKCHAIN :- A completely **open network** where **anyone can join, verify transactions, and view the ledger.** *Example: Bitcoin*

2. PRIVATE BLOCKCHAIN :- A **restricted blockchain managed by a single organization,** mainly used for internal business operations.

3. CONSORTIUM (FEDERATED) BLOCKCHAIN :- A blockchain **managed by multiple authorized organizations** (*instead of one company*), commonly used in banking and supply chain industries.

APPLICATIONS

- **Cryptocurrencies:** Enables secure peer-to-peer digital currency transactions.
- **Decentralized Finance (DeFi):** Supports lending, borrowing, and trading through smart contracts without traditional intermediaries.
- **Enterprise Solutions:** Used for digital identity, supply chain management, healthcare records, and secure institutional data sharing.

4. SECURITY AUDITING ! :-

MODULE 9 – CYBER CRIMES

1. CYBER CRIME ! :-

Cybercrime can be defined as an **act of committing any crime through a computer network against organizations** like government or private and even on any individual.

CATEGORIES OF CYBERCRIMES

1. **Individual Cyber Crimes** : Target individuals through phishing, spoofing, spam, cyberstalking, and identity theft.
2. **Organization Cyber Crimes** : Target businesses and institutions using malware, ransomware, hacking, and DDoS attacks.
3. **Property Cyber Crimes** : Target valuable assets such as credit cards, bank accounts, digital data, and intellectual property.
4. **Society Cyber Crimes** : Target society or national security through cyber-terrorism and attacks on critical infrastructure.

TYPES OF CYBERCRIMES

1. **Phishing**: **Fraudulent messages or emails tricking victims into revealing sensitive info** or downloading malwares. Attackers impersonate trusted entities to gain unauthorized access.
2. **Ransomware**: **Malware that locks a system or files**, demanding payment (usually in Bitcoin) to restore access. *Example: WannaCry.*
3. **DDoS Attack**: An attacker **floods a network with traffic to overload and crash** the target system, denying regular services.
4. **Malware**: **Malicious software** that disrupts or damages systems, including:
 - Trojans**: **Malware disguised as legitimate software**, providing backdoor access.
Example: Zeus.
 - Viruses**: **Malicious code** that damages files and system functions.
Example: Melissa.
 - Worms**: **Self-replicating malware** that spreads across networks.
Example: Ryuk, MyDoom.
5. **Man-in-the-Middle Attack**: The attacker **intercepts communications between two parties** to steal sensitive data, such as login credentials.

2. SCAMS AND FRAUDS : [EX : ⚡]

Internet fraud involves **using online services and software with internet access to do fraud** or take advantage of victims.

The term "internet fraud" generally covers cybercrime activity occurring over the internet or on email

Crimes like identity theft, phishing, and other activities designed to scam people out of money.

- ❖ **How it works**: Fraudsters use deceptive tactics such as **fake websites, phishing emails, and malicious links to trick users into revealing personal information**, downloading malware, or making fraudulent payments.

- ❖ **After-effects:** Financial loss, identity theft, emotional distress, damaged credit scores, and reduced trust in online interactions.
- ❖ **Prevention:** Avoid sharing personal information unnecessarily, use strong passwords, verify suspicious emails and websites, and keep software and antivirus programs updated..

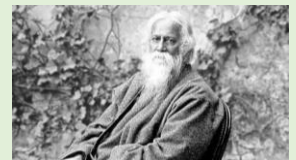
Email Phishing Scams  Fake emails trick you into clicking links or sharing data.	Greeting Card Scams  E-cards with malware infect your device when opened.	Credit Card Scams  Fake offers steal your card details for fraudulent use.	Online Dating Scams  Fake profiles build trust to ask for money or personal info.	Lottery Fee Fraud  You're told you won but must pay a fee to claim it.	The Nigerian Prince  Promised wealth in exchange for fees; money never comes.
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3. ANALYSIS OF CRIMES :-

When a cybersecurity breach occurs, digital forensic investigators analyze the incident to identify its cause and impact.

- Step 1 :- Evidence Collection:** Gather and preserve data from computers, servers, cloud storage, mobile devices, and system logs.
- Step 2 :- Root Cause Analysis:** Trace the attack to identify the entry point, attacker movement, and data theft path.
- Step 3 :- Attribution:** Analyze IP addresses and attacker behavior to determine whether the attack was carried out by insiders, cybercriminals, or state-sponsored groups.

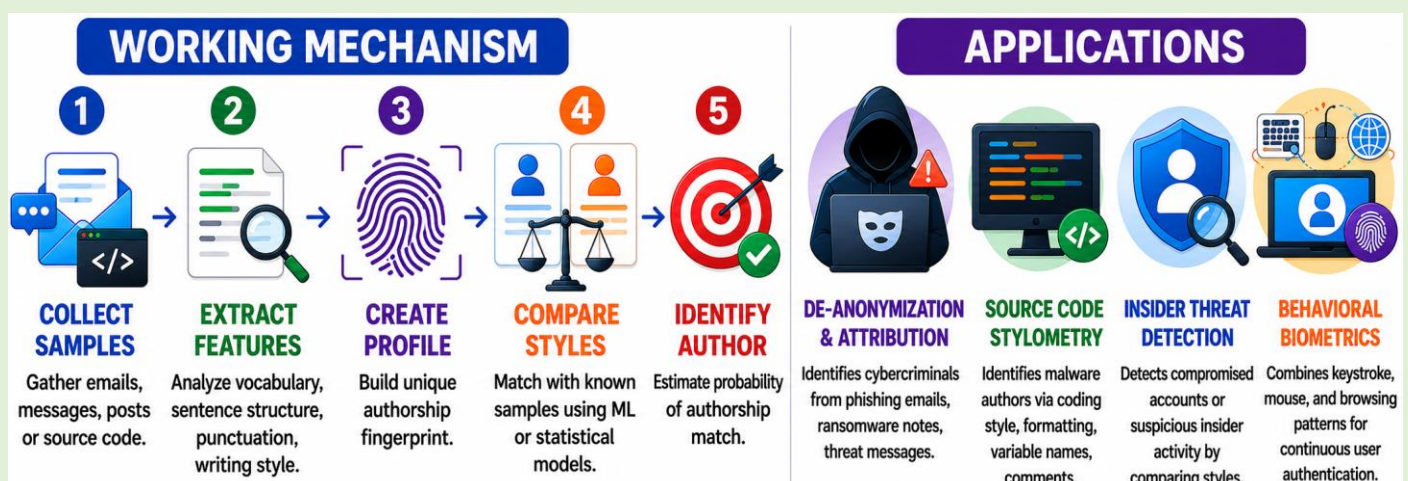
4. HUMAN BEHAVIOUR :-



5. STYLOMETRY :-

Stylometry is the statistical analysis of linguistic style (*word choice, sentence structure, punctuation habits*) used to identify authors , detect plagiarism, flag text generated by LLMs , etc

*Acts as a **digital fingerprint** to identify anonymous attackers, attribute cyberattacks, and detect insider threats.*



6. INCIDENT HANDLING:-

Incident Handling is the **organized approach/process for managing and responding to a security breach or cyber attack to limit damage** and reduce recovery time/cost.

Faster incident handling = lower financial/reputational damage.

TOOLS USED

- ✓ **SIEM:** Centralized log monitoring, threat detection, and incident investigation.
- ✓ **EDR & IDS/IPS:** Detect, monitor, contain, and block endpoint or network attacks.
- ✓ **Threat Intelligence & Ticketing:** Provide attack intelligence and track incident response.

IMPORTANCE

- ✓ **Limits Damage:** **Reduces attack impact**, downtime, and data loss through rapid response.
- ✓ **Ensures Compliance:** **Meets standards** such as **ISO 27001, HIPAA, GDPR, and NIST**.
- ✓ **Strengthens Security:** Lessons from incidents **improve future defenses** and organizational resilience.



7. INVESTIGATION METHODS:-

Cybersecurity investigations rely on a **combination of digital forensics, network analysis, and threat hunting** to **identify, contain, and analyze malicious activity**.

Investigators utilize these techniques to recover compromised data, reconstruct attack timelines, and trace the origin of cyber threats

METHODS

1. **Digital Forensics:** **Collects and analyzes digital evidence**, including deleted files, memory (RAM), and attack timelines.
2. **Network Analysis:** **Examines network traffic to detect unauthorized access**, data transfers, and attacker communications.
3. **Malware Analysis:** **Studies malware** using **static** (without execution) and **dynamic** analysis.
4. **Log & SIEM Analysis:** **Reviews system and security logs** to identify Indicators of Compromise (IoCs) and suspicious activities.
5. **Threat Hunting:** **Proactively searches networks for hidden threats** missed by automated security tools.

8. CRIMINAL PROFILING:-

It refers to building a **behavioral and technical profile of a likely suspect based on attack patterns**, tools used, timing, and target selection — **helps narrow down suspect pool.**

Often combined with **Cyber Trails** and log analysis to build a complete picture of the attacker.

1. **MODUS OPERANDI** : Identifies attack methods, tools, techniques that act as a digital fingerprint.
2. **PSYCHOLINGUISTICS**: Analyzes language and writing style in phishing emails, ransomware notes, or forum posts to infer origin, education, or intent.
3. **VICTIMOLOGY** : Studies targeted victims to determine the attacker's motive (*e.g. financial gain*).
4. **BEHAVIORAL INDICATORS**: Examines attack timing, activity patterns, and navigation habits to support attribution.
5. **MITRE ATT&CK FRAMEWORK**: Maps attacker tactics and techniques (TTPs) for threat profiling and analysis.

9. CYBER TRAILS:-

A cyber trail (*digital footprint*) **is the trace of data left behind by a user's online activity** (*IP logs, timestamps, metadata, cookies, transaction records*) that **investigators follow to trace the source** of a cyber crime.

FOLLOWING A CYBER TRAIL



Cyber trails can be hidden using VPN/proxy — investigators use advanced correlation techniques to de-anonymize.



MODULE 10 – DIGITAL FORENSICS

Digital Forensics is the **process of identifying, preserving, collecting, analyzing digital evidence from electronic devices to reconstruct cyberattacks** or investigate crimes

They are presented in a way that is **legally admissible in a court of law**

The 4 pillars of digital forensics are Identification, Preservation, Analysis, and Presentation

1. HISTORY OF DIGITAL FORENSICS

- ✓ **1980S:** Rise of personal computers → early computer crime cases begin.
- ✓ **1984:** FBI forms Computer Analysis and Response Team (CART).
- ✓ **1990S:** Formal methodologies developed; growth of internet crime.
- ✓ **2000S ONWARD:** Explosion of mobile/cloud forensics due to smartphones & cloud storage

2. BRANCHES OF DIGITAL FORENSICS

1. COMPUTER FORENSICS

Investigates computers, laptops, hard drives, and storage devices to recover digital evidence and reconstruct user activities. *Example: Recovering deleted finance records from a suspect's computer*

- **Recovers:** Deleted files, documents, emails, browser history, system logs.
- **Examines:** Hidden folders, OS artifacts, user activity. .

2. NETWORK FORENSICS

Analyzes network traffic to detect cyberattacks, unauthorized access, and suspicious communications. *Example: Tracing the source of a DDoS attack*

- **Examines:** Data packets, IP addresses, network logs.
- **Monitors:** Firewall logs, intrusion alerts, traffic patterns. .

3. MOBILE FORENSICS

Extracts and analyzes evidence from smartphones, tablets, SIM cards, and memory cards.

Example: Recovering deleted WhatsApp messages

- **Recovers:** Call logs, SMS, contacts, photos.
- **Extracts:** App messages, location history, device data. .

4. DATABASE FORENSICS

Investigates databases to identify unauthorized access, data modification, or suspicious transactions. *Example: Identifying who modified a banking transaction*

- **Examines:** Database records, transactions, user activities.
- **Detects:** Deleted records, data tampering, unauthorized changes. .

5. EMAIL FORENSICS

Analyzes emails to determine authenticity, sender identity, and malicious activity. *Example: Tracing the sender of a phishing email.*

- **Examines:** Email headers, content, attachments.
- **Detects:** Phishing, spoofing, malicious attachments.

7. CLOUD FORENSICS

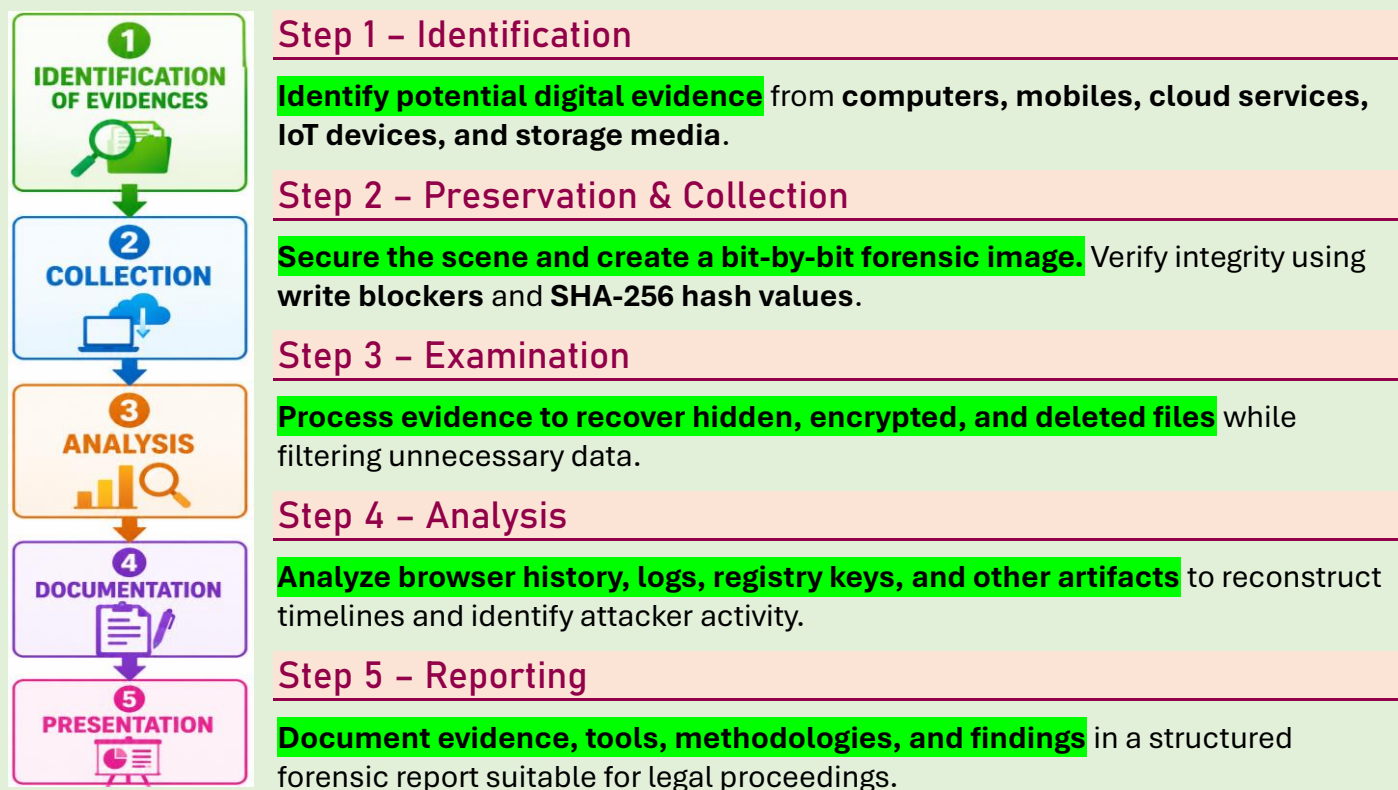
Investigates cyber incidents involving cloud platforms, virtual machines, and online storage.

Example: Investigating a cloud account compromise

- **Examines:** Cloud logs, virtual machines, cloud storage.
- **Investigates:** Unauthorized access, data breaches, account misuse.

 COMPUTER FORENSICS Investigates files and evidence from computers.	 DATABASE FORENSICS Examines databases for tampering and unauthorized access.	 MEDIA FORENSICS Examines digital images, videos, and audio evidence.
 NETWORK FORENSICS Analyzes network traffic to detect cyberattacks.	 CLOUD FORENSICS Investigates incidents involving cloud services and storage.	 SOFTWARE FORENSICS Investigates software code and application tampering.
 MOBILE FORENSICS Recovers evidence from smartphones and tablets.	 MEMORY (RAM) FORENSICS Analyzes volatile memory for active threats.	 CYBER FORENSICS Overall investigation of cybercrime using digital evidence.
 EMAIL FORENSICS Investigates phishing, spoofing, and email fraud.		
 MALWARE FORENSICS Analyzes malicious software and its behavior.		

3. DIGITAL FORENSIC INVESTIGATION METHODS



4. REPORTING

Forensic reporting is the **documentation of the entire investigation process**, findings, methodology, and conclusions in a clear, structured, and legally defensible format.

- | | |
|--|---|
| 1. Case details & examiner info | 4. Tools & methodology used |
| 2. Objective/scope of investigation | 5. Findings/analysis |
| 3. Evidence details (hash values, acquisition method) | 6. Conclusion |
| | 7. Supporting exhibits (screenshots, logs) |

Points to Remember: Report must be clear enough for a non-technical judge/jury to understand; must be reproducible by another examiner.

5. MANAGEMENT OF EVIDENCE

How is the collected evidence protected and handled ?

It is the process of **collecting, preserving, storing, documenting, transporting, and presenting digital evidence** while maintaining its integrity and admissibility in court.

OBJECTIVES

- | | |
|--------------------------------|-------------------------------|
| a) Preserve evidence integrity | c) Maintain chain of custody |
| b) Prevent tampering | d) Ensure legal admissibility |

STEPS IN EVIDENCE MANAGEMENT

- 1. IDENTIFICATION :- Locate potential digital evidence.** *Examples: Laptop, mobile, cloud data.*
- 2. COLLECTION :- Collect evidence** using forensic procedures. Create forensic image instead of working on original.
- 3. PRESERVATION :- Prevent modification of evidence.** Use write blockers and hash values (SHA-256).
- 4. DOCUMENTATION :- Record evidence details.** Maintain date, time, owner, investigator.
- 5. SECURE STORAGE :- Store evidence securely.** Use sealed evidence bags & secure storages.
- 6. CHAIN OF CUSTODY :-** Maintain complete **record of every person handling evidence.**
- 7. PRESENTATION :- Present verified evidence in court** with proper documentation.

CHAIN OF CUSTODY

It is the **documented record of every person who collects, handles , access the evidence** from collection until court presentation.

Ensures evidence integrity ; Prevents tampering ; Maintains legal admissibility

6. CHALLENGES OF DIGITAL FORENSICS

- 1) Data Volume & Complexity** – Huge amount of digital data makes evidence difficult to locate.
- 2) Encryption & Data Protection** – Encrypted devices and apps prevent access without decryption keys.
- 3) Volatile Evidence & Data Loss** – RAM and temporary data can disappear if not collected immediately. *(Live acquisition is important.)*
- 4) Cloud Computing & Jurisdiction** – Data stored across countries creates legal and jurisdictional issues.
- 5) Legal Admissibility & Standards** – Evidence must follow proper forensic procedures and maintain the **Chain of Custody** to be accepted in court.

MODULE 11 – CYBER LAW_{DA}

1. WHAT IS CYBER LAW?

Cyber law refers to the **legal rules that govern online activities**, covering areas like cybercrimes (hacking, identity theft, fraud), intellectual property, data protection, privacy, e-commerce, and digital contracts.

It aims to **protect individuals and organizations from cyber threats** and ensure safe, ethical use of digital technologies.

TYPES OF CYBER CRIME

- 1. HACKING:** Unauthorized access to computer systems to steal, alter, or destroy data.
- 2. DATA THEFT:** Stealing sensitive personal or corporate data.
- 3. IDENTITY THEFT (INCLUDING PASSWORD THEFT):** Fraudulent use of someone's personal information for malicious purposes.
- 4. EMAIL SPOOFING:** Sending emails with a forged sender address to deceive recipients.
- 5. SENDING OFFENSIVE MESSAGES:** Using online platforms to send threatening, harassing, or offensive messages.
- 6. VOYEURISM:** Recording or disseminating private or sexual images without consent
- 7. CYBERTERRORISM:** Using internet to cause fear, harm, or disruption for social motives.

2. WHAT IS CYBER TERRORISM?

Cyber terrorism involves using **digital methods to create violence** or to harm, disrupt, or instil fear.

It usually pertains to attacks that are driven by political or ideological beliefs, aiming at computer systems, networks, or the **structures that hold information**

METHODS: Hacking, Malware, Denial of Service (DoS), Data Breaches, Cyber Espionage.

TARGETS:-

- ❖ **CRITICAL INFRASTRUCTURE:** Key systems like **power grids**, water supplies, etc.
- ❖ **GOVERNMENT:** Attacks on government operations and **national security**.
- ❖ **PRIVATE SECTOR:** Attacks on **financial institutions** and businesses .

3. CYBERCRIMES & OFFENCES UNDER IPC :-

IPC Sections:

SECTION	DEALS WITH	PUNISHMENT / KEY POINT
SEC 292	Obscene material	1st Offence: 2 years + ₹2,000 fine Repeat: 5 years + ₹5,000 fine
SEC 354C	Voyeurism	1st Offence: 1–3 years; Repeat: 3–7 years
SEC 379	Data Theft	Up to 3 years imprisonment or fine
SEC 420	Cheating / Online Fraud	Up to 7 years imprisonment and fine

IT ACT 2000 Sections:

SECTION	TOPIC	PUNISHMENTS
Sec 43	Unauthorized system/data access	Compensation (Civil Liability)
Sec 65	Tampering with source code	Up to 3 years jail or ₹2 lakh fine , or both
Sec 66	Computer offences (Hacking)	Up to 3 years jail or ₹5 lakh fine , or both
Sec 66C	Identity / digital sign theft	up to 3 years jail + ₹1 lakh fine
Sec 66D	Cheating by personation using computer resources	Up to 3 years + ₹1 lakh fine
Sec 66E	Violation of privacy	3 years jail or ₹2 lakh fine
Sec 66F	Cyber Terrorism	Life Imprisonment
Sec 67	Publishing or transmitting obscene material	1st Offence: 3 years + ₹5 lakh fine
Sec 69 	Govt's power to monitor, or decrypt information	National Security
Sec 72	Breach of confidentiality and privacy	Up to 2 years jail or ₹1 lakh fine , or both

2008 Amendment:

- Added **Sec 43A** (Data Protection).
- Added **Sec 66C–66F** (Identity Theft, Online Fraud, Privacy, Cyber Terrorism).
- Added **Sec 67A & 67B** (Sexually Explicit Content & Child Pornography).
- **Sec 66A** was **struck down in 2015** (*Shreya Singhal case*).

4. WHAT IS HACKTIVISM?

Hactivism combines **hacking** and **activism**. It involves breaking into computer systems to **promote political or social causes**.

The individual who performs an act of hacktivism is said to be a **hacktivist**. He may **deface websites, leak data, or launch cyberattacks to protest about issues**.

Main goal is to send a message and raise awareness for the cause they are promoting.

Example: In 2008, the group **Anonymous** attacked the **Church of Scientology** to protest censorship by taking down its websites and leaking documents.

5. EVIDENTIARY VALUE OF EMAIL/SMS

Refers to the **legal admissibility and weight given to electronic communications** (emails, SMS, WhatsApp messages) as evidence in court proceedings.

LEGAL PROSPECTS :-

- ❖ **Section 65B of the Indian Evidence Act, 1872:** Electronic records are admissible in court only when accompanied by a **Section 65B certificate** confirming their authenticity.
- ❖ **Anvar P.V. v. P.K. Basheer (2014):** The Supreme Court ruled that a **Section 65B certificate** is **mandatory** for electronic evidence to be admissible

Email headers (metadata) help prove origin/authenticity. - Must maintain chain of custody (see Unit 10) for such evidence to hold up.

6. RBI ACT ; INTELLECTUAL PROPERTY RIGHTS ACT IN INDIA

RBI Act :-

- Governs **digital banking transactions**.
- Prescribes security standards for **UPI, mobile banking, and payment systems**.
- Includes **NPCI guidelines**.
- Helps prevent and penalize **cyber fraud in financial transactions**.

IPR (Intellectual Property Rights) :-

- Protects **software, digital content, trademarks, and copyrights**.
- Prevents **online piracy and infringement**.
- Covered under the **Copyright Act, 1957 (as amended)** and the **Trademarks Act** (including domain name and brand misuse).

7. JURISDICTION OF CYBER CRIME

Jurisdiction **refers to the legal authority of a court or country to try a cyber crime case**. It is a major challenge because cyber crimes are borderless.

Challenges:

- 1) **Cross-Border Involvement:** Attacker, victim, and server **may be in different countries**.
- 2) **Different Cyber Laws:** **Each country has its own cyber laws** and regulations.
- 3) **Extradition Issues:** **Difficulty in bringing cyber criminals** to the concerned country for trial.
- 4) **Cloud Jurisdiction:** Data may be stored on cloud servers across multiple jurisdictions.

Basis for Jurisdiction under the IT Act:

- **Section 75 of the IT Act** provides **extra-territorial application**.

It applies to offences committed **outside India** if they involve a **computer, computer system, or network located in India**.

8. Cyber Security Awareness Tips

1. **STRONG PASSWORDS & MFA:** Use strong, unique passwords and enable **2FA/MFA**.
2. **REGULAR UPDATES:** Keep the operating system, software, and antivirus updated.
3. **SECURE BROWSING:** Use **HTTPS** websites and avoid public Wi-Fi for sensitive transactions (or use a **VPN**).
4. **VERIFY THE SENDER:** Confirm the sender before sharing sensitive information through email, SMS, or calls.
5. **USE GENUINE SOFTWARE:** Install only licensed and genuine software.
6. **SECURE DEVICES:** Lock devices using a PIN, password, or biometric authentication.
7. **REPORT CYBER CRIME:** Report suspicious activities immediately through **cybercrime.gov.in** or the **Cyber Crime Helpline (1930)**.

COPYRIGHTS



Protects
creative works



Covers books, music,
software, art



Arises automatically
on creation



Prevents copying
of the work



Example: Novel,
movie, app code

PATENTS



Protects
new inventions



Covers products,
processes, designs



Requires official
registration



Prevents making,
using, or selling
the invention



Example: New engine,
medical device

TRADEMARKS



Protects
brand identity



Covers names,
logos, slogans



Usually requires
registration



Prevents brand misuse
or confusion



Example: Nike logo,
Coca-Cola name